

A Magician, a Coin, and Log-Periodic Hitting Probabilities

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Abstract

This problem was inspired by an audience-participation routine performed by magician Wes Iseli. Audience members predict successive tosses of a coin: correct predictors remain standing and incorrect predictors sit down. In the performance, the final two participants are instructed to make different choices, guaranteeing a single survivor; the magician then reveals a written prediction describing that person. Removing the final coordination gives a natural family of questions. Let $H_{n,k}$ be the probability that a fair, independent thinning process begun with n people ever has exactly k survivors; the original one-survivor probability is $P_n = H_{n,1}$. We derive exact finite and infinite formulas for every k , and show that for each fixed k ,

$$H_{n,k} = F_k(\{\log_2 n\}) + o(1),$$

where F_k is a smooth, nonconstant function of period one. Thus none of the sequences $(H_{n,k})_{n \geq k}$ converges, although $H_{2n,k} - H_{n,k} \rightarrow 0$. The case $k = 2$ explains the point at which the stage routine imposes opposite choices. We also study the dependence on k : its logarithm gives an exact moving phase in a recentered formula, but the profile also narrows. In particular,

$$\max_{\theta} F_k(\theta) \sim (2\pi k)^{-1/2}, \quad \int_0^1 F_k(\theta) d\theta \sim (k \ln 2)^{-1}.$$

After centering at $\{\log_2 k\}$ and scaling phase by $\sqrt{k}$, the profiles approach a Gaussian.

1 The magician's problem

1.1 The performance that motivated the question

The starting point is a routine performed by Wes Iseli on *Penn & Teller: Fool Us*, in the broadcast episode “Fool Us: The Home Game” [1, 2]. Iseli gives the host a written prediction, asks the audience to stand, and runs a large game of heads and tails. Each participant signals a choice; Iseli tosses a half-dollar; and those whose choice does not match the outcome sit down. The process is repeated, rapidly reducing the standing audience. Penn Jillette later described the routine as Iseli’s whole-audience coin-flip effect [3].

The stage routine contains a crucial piece of management. When only two audience members remain, Iseli tells them to choose different sides. The next toss must therefore leave exactly one of them standing. The written prediction is then opened and found to describe visible features of that final participant.

Thus the performance itself does not rely on chance to avoid the zero-survivor outcome at the final step: the instruction to the last two participants prevents it. The mathematical question arises by removing that safeguard.

1.2 The probabilistic variant

Imagine that every member of an audience of size n begins standing and independently predicts the result of a fair coin toss. Those who predict correctly remain standing; the others sit down permanently. The remaining participants make fresh independent predictions, and the experiment is repeated with no coordination, including when only two or a few people remain.

We assume throughout that all predictions are independent fair choices and that fresh independent choices are made in every round. The outcome of the coin is common to the audience, but this does not create dependence among the correctness indicators. Conditional on either outcome of the coin, each person is correct independently with probability $1/2$, and the conditional joint distribution is the same for heads and tails. Consequently, each round independently retains every active participant with probability $1/2$.

Let N_t be the number standing after t rounds, with $N_0 = n$. For $1 \leq k \leq n$, define

$$H_{n,k} = \mathbb{P}_n\{N_t = k \text{ for some } t \geq 0\}. \quad (1)$$

Equivalently, one may stop the process the first time it reaches k ; stopping does not change the probability of reaching that state. We put $H_{k,k} = 1$ and $H_{n,k} = 0$ for $n < k$. The original question is

$$P_n = H_{n,1},$$

the probability that exactly one person remains before the chain jumps to zero. The probability $H_{n,2}$ instead measures whether the uncoordinated process ever presents the magician with a final pair. What the stage routine does after reaching that pair is a separate, deterministic intervention.

2 Geometric lifetimes and exact hitting probabilities

For participant i , let L_i be the number of consecutive correct predictions made before the first incorrect prediction. Then the L_i are independent and

$$\mathbb{P}(L_i = \ell) = 2^{-(\ell+1)}, \quad \mathbb{P}(L_i < \ell) = 1 - 2^{-\ell}, \quad \ell = 0, 1, 2, \dots$$

Moreover,

$$N_t = \#\{i : L_i \geq t\}.$$

If $L_{[1]} \geq \dots \geq L_{[n]}$ are the lifetimes in decreasing order, then, for $n > k$, the chain reaches k exactly when

$$L_{[k]} > L_{[k+1]}. \quad (2)$$

For $k = 1$ this says that the magician's probabilistic version succeeds exactly when the maximum lifetime is unique.

2.1 An occupation formula for every target size

Theorem 2.1 (Exact formulas). *For $n \geq k \geq 1$,*

$$\begin{aligned} H_{n,k} &= (1 - 2^{-k}) \sum_{t=0}^{\infty} \mathbb{P}\{\text{Binomial}(n, 2^{-t}) = k\} \\ &= (1 - 2^{-k}) \binom{n}{k} \sum_{t=0}^{\infty} 2^{-kt} (1 - 2^{-t})^{n-k}. \end{aligned} \quad (3)$$

It also has the finite rational form

$$H_{n,k} = (1 - 2^{-k}) \binom{n}{k} \sum_{r=0}^{n-k} (-1)^r \binom{n-k}{r} \frac{2^{k+r}}{2^{k+r} - 1}. \quad (4)$$

When $n > k$, the last fraction may be replaced by $1/(2^{k+r} - 1)$.

Proof. Let

$$V_k = \sum_{t \geq 0} \mathbf{1}_{\{N_t = k\}}$$

be the number of visits to k . Once the chain leaves k it can never return. While at k , it stays there for another round exactly when all k people survive, an event of probability 2^{-k} . Conditional on reaching k , V_k therefore has geometric mean $(1 - 2^{-k})^{-1}$. Hence

$$\mathbb{E}V_k = \frac{H_{n,k}}{1 - 2^{-k}}.$$

On the other hand, each original participant survives t rounds with probability 2^{-t} , independently, so

$$N_t \sim \text{Binomial}(n, 2^{-t}).$$

Summing $\mathbb{P}(N_t = k)$ over t proves (3).

For the finite form, expand $(1 - 2^{-t})^{n-k}$ and sum the resulting geometric series from $t = 0$:

$$\sum_{t=0}^{\infty} 2^{-(k+r)t} = \frac{2^{k+r}}{2^{k+r} - 1}.$$

If $n > k$, the constant part in $2^{k+r}/(2^{k+r} - 1) = 1 + 1/(2^{k+r} - 1)$ cancels by the binomial theorem. $\square$

For $k = 1$, Theorem 2.1 recovers the original formula

$$P_n = \frac{n}{2} \sum_{t=1}^{\infty} 2^{-t} (1 - 2^{-t})^{n-1}, \quad n \geq 2, \quad (5)$$

and the finite expression

$$P_n = \frac{n}{2} \sum_{r=0}^{n-1} (-1)^r \binom{n-1}{r} \frac{1}{2^{r+1} - 1}, \quad n \geq 2. \quad (6)$$

In particular, all the hitting probabilities are rational for finite n .

Conditioning on the first round gives another useful exact description. For $n > k$,

$$H_{n,k} = \frac{1}{2^n - 1} \sum_{j=k}^{n-1} \binom{n}{j} H_{j,k}. \quad (7)$$

The denominator removes the self-loop in which all n people survive. For example,

$$P_2 = \frac{2}{3}, \quad P_3 = \frac{5}{7}, \quad P_4 = \frac{76}{105}.$$

2.2 The case of two survivors and the televised finale

Taking $k = 2$ gives

$$H_{n,2} = \frac{3}{4} \binom{n}{2} \sum_{t=0}^{\infty} 4^{-t} (1 - 2^{-t})^{n-2}. \quad (8)$$

The first few values are

$$H_{2,2} = 1, \quad H_{3,2} = \frac{3}{7}, \quad H_{4,2} = \frac{18}{35}.$$

This is the probability that independent play ever produces the pair required for the staged ending. Once the show reaches two, opposite choices force one survivor on the next toss. If those two instead continue independently, the chance of ever reaching one is only

$$P_2 = \frac{2}{3}.$$

Indeed, from two people a round leaves 0, 1, 2 survivors with probabilities $1/4, 1/2, 1/4$; conditional on leaving the self-loop at 2, the chance of moving to 1 is $(1/2)/(3/4) = 2/3$.

2.3 Connection with tied geometric maxima

Let $W_{n,k}$ be the probability that exactly k of the lifetimes $L_1, \dots, L_n$ share their maximum. After the chain first reaches k , the maximum has multiplicity k precisely when the first non-self transition is the jump from k to zero. The conditional probability of that jump is

$$\frac{2^{-k}}{1 - 2^{-k}} = \frac{1}{2^k - 1}.$$

Consequently,

$$H_{n,k} = (2^k - 1)W_{n,k}. \quad (9)$$

Since exactly one multiplicity occurs, this also gives the identity

$$\sum_{k=1}^n \frac{H_{n,k}}{2^k - 1} = 1. \quad (10)$$

3 What the law of large numbers predicts

At every deterministic round,

$$N_t \sim \text{Binomial}(n, 2^{-t}), \quad \mathbb{E}N_t = n2^{-t}. \quad (11)$$

Thus the population is approximately halved at every round while it remains large.

This can be made precise even when the number of rounds grows with n . Fix $\varepsilon > 0$ and put

$$t_- = \lfloor (1 - \varepsilon) \log_2 n \rfloor.$$

Then $\mathbb{E}N_{t_-} \asymp n^\varepsilon \rightarrow \infty$ and

$$\frac{\text{Var}(N_{t_-})}{(\mathbb{E}N_{t_-})^2} \leq \frac{1}{\mathbb{E}N_{t_-}} \rightarrow 0.$$

Chebyshev's inequality therefore gives

$$\frac{N_{t_-}}{\mathbb{E}N_{t_-}} \xrightarrow{\mathbb{P}} 1.$$

Many participants remain at this time. On the other hand, for

$$t_+ = \lceil (1 + \varepsilon) \log_2 n \rceil,$$

we have $\mathbb{E}N_{t_+} \rightarrow 0$, and Markov's inequality gives

$$\mathbb{P}(N_{t_+} \geq 1) \leq \mathbb{E}N_{t_+} \rightarrow 0.$$

The decisive rounds therefore lie near $\log_2 n$.

This immediately suggests a scaling principle. If the audience size is doubled, one initial halving should return it to approximately its former size. Indeed, if B_n is the number surviving the first round from $2n$ people, then

$$B_n \sim \text{Binomial}(2n, 1/2), \quad \frac{B_n}{n} \xrightarrow{\mathbb{P}} 1,$$

and the Markov property gives

$$H_{2n,k} = \mathbb{E}[H_{B_n,k}]. \quad (12)$$

For every fixed target k , one is therefore led to guess

$$H_{2n,k} - H_{n,k} \rightarrow 0. \quad (13)$$

There is an important logical subtlety here. The law of large numbers says that B_n/n is close to one; it does not by itself say that P_{B_n} is close to P_n , or more generally that $H_{B_n,k}$ is close to $H_{n,k}$. That conclusion also requires local regularity of the hitting probabilities. The analysis below supplies exactly that regularity and proves (13).

The law of large numbers also stops being informative inside the critical window. If

$$n = 2^{m+\theta}, \quad 0 \leq \theta < 1,$$

then after $m + j$ rounds the expected number standing is

$$\mathbb{E}N_{m+j} = 2^{\theta-j},$$

which is of order one. There is no concentration in this regime. Instead, the relevant binomial variables have Poisson limits. The fractional part $\theta = \{\log_2 n\}$ is not washed away, and it is the source of the oscillation.

4 The log-periodic limiting profiles

For each target $k \geq 1$, define

$$F_k(\theta) = \frac{1 - 2^{-k}}{k!} \sum_{j \in \mathbb{Z}} (2^{\theta-j})^k \exp(-2^{\theta-j}), \quad \theta \in \mathbb{R}. \quad (14)$$

The series and all its termwise derivatives converge locally uniformly, so F_k is real analytic. Shifting the summation index gives

$$F_k(\theta + 1) = F_k(\theta). \quad (15)$$

We write $F = F_1$ for the profile of the original problem.

Theorem 4.1 (Uniform log-periodic approximation). *For every fixed $k \geq 1$,*

$$\sup_{2^m \leq n < 2^{m+1}} |H_{n,k} - F_k(\{\log_2 n\})| \longrightarrow 0 \quad (m \rightarrow \infty). \quad (16)$$

In particular, $H_{n,k} = F_k(\{\log_2 n\}) + o(1)$ as $n \rightarrow \infty$.

Proof. Write

$$m = \lfloor \log_2 n \rfloor, \quad \theta = \log_2 n - m,$$

put $t = m + j$ in (3), and set $x_j = 2^{\theta-j} = n2^{-t}$. Then

$$H_{n,k} = (1 - 2^{-k}) \sum_{j \geq -m} \binom{n}{k} \left(\frac{x_j}{n}\right)^k \left(1 - \frac{x_j}{n}\right)^{n-k}. \quad (17)$$

For fixed x and fixed k , the binomial mass in the sum tends to $x^k e^{-x}/k!$. This convergence is uniform for x in any compact subinterval of $(0, \infty)$.

It remains to control both tails uniformly in $0 \leq \theta < 1$. For the small dyadic points, the elementary bound $\binom{n}{k} \leq n^k/k!$ gives

$$\sum_{x_j \leq \delta} \binom{n}{k} (x_j/n)^k = O_k(\delta^k).$$

For the large points, when $n \geq 2k$ and $t \geq 1$,

$$\binom{n}{k} \left(\frac{x}{n}\right)^k \left(1 - \frac{x}{n}\right)^{n-k} \leq \frac{x^k}{k!} e^{-x/2}.$$

The dyadic sum of the right-hand side over $x \geq A$ tends uniformly to zero as $A \rightarrow \infty$; the $t = 0$ term vanishes when $n > k$. The omitted terms in the two-sided limiting series have the still smaller tail $x^k e^{-x}/k!$. First let $n \rightarrow \infty$, then $\delta \downarrow 0$ and $A \rightarrow \infty$. This proves (16). $\square$

The theorem confirms the scaling prediction, simultaneously for every fixed target:

Corollary 4.2. *For each fixed $k \geq 1$,*

$$H_{2n,k} - H_{n,k} \longrightarrow 0.$$

Proof. The integers n and $2n$ have the same logarithmic phase. Apply Theorem 4.1 to both terms. $\square$

This result also completes the law-of-large-numbers argument. If $r_n/n \rightarrow 1$, then the uniform approximation and continuity of F_k give $H_{r_n,k} - H_{n,k} \rightarrow 0$. Since $B_n/n \rightarrow 1$ in probability, it follows that $H_{B_n,k} - H_{n,k} \rightarrow 0$ in probability. The difference is bounded by one, so this convergence also holds in L^1 ; taking expectations in (12) yields the same doubling relation. Thus the law of large numbers provides the right coupling and intuition; the log-periodic estimate supplies the local regularity that the law alone does not contain.

5 Fourier analysis and nonconvergence

Put $L = \ln 2$ and $\tau = 2\pi/L$. The function F_k is the periodization of

$$h_k(u) = \frac{1 - 2^{-k}}{k!} 2^{ku} \exp(-2^u).$$

With the convention $\hat{F}_k(\ell) = \int_0^1 F_k(\theta) e^{-2\pi i \ell \theta} d\theta$, the substitution $x = 2^u$ gives

$$\hat{F}_k(\ell) = \frac{1 - 2^{-k}}{k!L} \Gamma(k - i\tau\ell), \quad \ell \in \mathbb{Z}. \quad (18)$$

In particular, the phase average is

$$\mu_k := \int_0^1 F_k(\theta) d\theta = \frac{1 - 2^{-k}}{kL}. \quad (19)$$

For example, the uniform logarithmic-phase average of the large-audience probability of ever seeing two survivors is

$$\mu_2 = \frac{3}{8 \ln 2} = 0.5410106403 \dots$$

Periodicity alone would not rule out convergence if F_k were constant. However, the gamma function has no zeros, so $\hat{F}_k(1) \neq 0$ for every k . Every profile F_k is therefore nonconstant.

Theorem 5.1 (Nonconvergence for every fixed target). *For every fixed $k \geq 1$, the sequence $(H_{n,k})_{n \geq k}$ does not converge. More precisely, for every $\theta \in [0, 1)$ there are integers $n_m \rightarrow \infty$ such that*

$$H_{n_m, k} \longrightarrow F_k(\theta).$$

Proof. For $0 < \theta < 1$, take $n_m = \lfloor 2^{m+\theta} \rfloor$; for $\theta = 0$, take $n_m = 2^m$. Then $\{\log_2 n_m\} \rightarrow \theta$. Theorem 4.1 and continuity of F_k give the stated limit. Since F_k is nonconstant, two phases give distinct subsequential limits. $\square$

For the original problem, for example,

$$\lim_{m \rightarrow \infty} P_{2^m} = F_1(0) = 0.721352103336862 \dots,$$

whereas

$$\lim_{m \rightarrow \infty} P_{\lfloor 2^{m+1/2} \rfloor} = F_1(1/2) = 0.721342937541388 \dots$$

These two limits differ by only about 9.17×10^{-6} , but they are not equal.

6 Why the one-survivor oscillation is almost a cosine

Equation (18) with $k = 1$ gives the complete Fourier expansion

$$F_1(\theta) = \frac{1}{2L} + \frac{1}{L} \operatorname{Re} \sum_{\ell=1}^{\infty} \Gamma(1 - i\tau\ell) e^{2\pi i \ell \theta}. \quad (20)$$

Its mean is

$$\mu_1 = \frac{1}{2 \ln 2} = 0.721347520444482 \dots$$

Let $A_\ell = 2|\widehat{F}_1(\ell)|$ be the amplitude of the ℓ th real harmonic. The identity

$$|\Gamma(1 + iy)|^2 = \frac{\pi y}{\sinh(\pi y)}$$

gives

$$A_\ell = \frac{1}{L} \sqrt{\frac{\pi \tau \ell}{\sinh(\pi \tau \ell)}}. \quad (21)$$

The first amplitudes are

$$A_1 = 7.1301172597 \times 10^{-6}, \quad A_2 = 6.6033857077 \times 10^{-12}, \quad A_3 = 5.2962355285 \times 10^{-18}.$$

Every harmonic after the first is negligible on the scale of the first one. Consequently,

$$F_1(\theta) \approx \frac{1}{2 \ln 2} + 7.13011726 \times 10^{-6} \cos(2\pi(\theta - 0.86110373)).$$

Numerically,

$$\begin{aligned} \min F_1 &= 0.721340390331955 \dots, \\ \max F_1 &= 0.721354650566475 \dots \end{aligned}$$

The peak-to-peak oscillation is approximately 1.4260×10^{-5} . This explains why numerical experiments can easily suggest convergence to $1/(2 \ln 2)$ even though no limit exists.

Analytically, F_1 is obtained by adding all integer translates of the smooth bump h_1 . The translates overlap almost uniformly. Equivalently, the Fourier transform of h_1 is sampled at the relatively high frequencies $2\pi\ell$. Formula (21) shows exponential decay, asymptotically of size

$$A_\ell \asymp \sqrt{\ell} \exp\left(-\frac{\pi^2}{\ln 2} \ell\right).$$

7 Changing the target size: phase and shape

The graphs of F_k suggest that increasing k mainly changes a phase. There is a genuine phase effect, but it is not an exact shape invariance. To separate vertical size from shape, normalize each profile to have mean one:

$$G_k(\theta) = \frac{F_k(\theta)}{\mu_k} = \frac{kL}{1 - 2^{-k}} F_k(\theta). \quad (22)$$

Equation (18) becomes

$$\widehat{G}_k(\ell) = \frac{\Gamma(k - i\tau\ell)}{\Gamma(k)}. \quad (23)$$

There is also a probabilistic interpretation. If X_k has the gamma density $x^{k-1}e^{-x}/\Gamma(k)$ on $(0, \infty)$, then G_k is exactly the density on the unit circle of the fractional part $\{\log_2 X_k\}$.

The gamma recurrence gives

$$\widehat{G}_{k+1}(\ell) = \left(1 - \frac{i\tau\ell}{k}\right) \widehat{G}_k(\ell),$$

or, equivalently, the exact differential recurrence

$$G_{k+1} = G_k - \frac{1}{kL} G'_k. \quad (24)$$

This formula explains the visual phase shift. If only the first harmonic is visible, write

$$G_k(\theta) = 1 + 2\rho_k \cos(2\pi(\theta - \varphi_k)) + \text{higher harmonics}, \quad (25)$$

where

$$\rho_k = \frac{|\Gamma(k - i\tau)|}{\Gamma(k)}, \quad \varphi_k = -\frac{\arg \Gamma(k - i\tau)}{2\pi}.$$

Then

$$\frac{\rho_{k+1}}{\rho_k} = \sqrt{1 + \frac{\tau^2}{k^2}}, \quad \varphi_{k+1} - \varphi_k = \frac{1}{2\pi} \arctan\left(\frac{\tau}{k}\right) \pmod{1}. \quad (26)$$

Thus, after subtracting the mean and ignoring the vertical amplitude, curves in the one-harmonic regime really do look like shifted copies of one cosine.

They are not, however, exact translated copies of F_1 . Translation preserves all Fourier magnitudes, whereas

$$\left| \frac{\widehat{G}_k(\ell)}{\widehat{G}_1(\ell)} \right| = \prod_{r=1}^{k-1} \sqrt{1 + \frac{\tau^2 \ell^2}{r^2}}, \quad (27)$$

which depends on the harmonic ℓ . No single vertical rescaling can remove that dependence. For large k and fixed ℓ ,

$$\widehat{G}_k(\ell) = \exp \left\{ -2\pi i \ell \log_2 k - \frac{2\pi^2 \ell^2}{kL^2} + \frac{i\tau \ell}{2k} + O_\ell(k^{-2}) \right\}. \quad (28)$$

The leading imaginary term is the moving phase (with a smaller circular-mean correction of order k^{-1}); the Gaussian damping of the harmonics is the changing shape.

8 The large-target profile

The limit in Theorem 4.1 first sends $n \rightarrow \infty$ with k fixed. We now study the resulting profile F_k as $k \rightarrow \infty$. These statements do not assert a joint limit for an arbitrary target $k = k(n)$.

The phase and the changing shape become especially transparent in an exact recentered formula. Put

$$\phi_k = \{\log_2 k\}, \quad u = \{\theta - \phi_k\}, \quad I(x) = x - 1 - \ln x,$$

and

$$A_k = (1 - 2^{-k}) \frac{k^k e^{-k}}{k!}. \quad (29)$$

Reindexing the defining sum for F_k gives the exact identity

$$F_k(\theta) = A_k \sum_{r \in \mathbb{Z}} \exp[-kI(2^{u-r})]. \quad (30)$$

This is the precise sense in which k acts as a phase: $\{\log_2 k\}$ is exactly the horizontal recentering, and doubling k leaves that phase unchanged. But k also multiplies the rate function I , so increasing k contracts the profile. The functions are not pure translations of F_1 .

Stirling's formula yields

$$A_k = \frac{1 - 2^{-k}}{\sqrt{2\pi k}} \left(1 - \frac{1}{12k} + O(k^{-2}) \right). \quad (31)$$

Since $I(x) \geq 0$ with equality only at $x = 1$, one dyadic term dominates when the phase is near ϕ_k . In particular, for some constant $c > 0$,

$$\max_{\theta} F_k(\theta) = A_k(1 + O(e^{-ck})) \sim \frac{1}{\sqrt{2\pi k}}. \quad (32)$$

The maximizing phase is $\theta_{\max,k} = \phi_k + O(e^{-ck}) \pmod{1}$; thus ϕ_k is the exact recentering parameter and the asymptotic peak phase, not necessarily the exact finite- k maximizer. The phrase “of order $1/\sqrt{k}$ ” therefore describes the maximum or envelope, not the value at every phase and not pointwise monotonicity in k . The peak itself moves as k changes. Its phase average, from (19), is smaller:

$$\mu_k \sim \frac{1}{k \ln 2}. \quad (33)$$

The local shape is universal. For each fixed real s , with the argument read modulo one,

$$\sqrt{2\pi k} F_k\left(\phi_k + \frac{s}{L\sqrt{k}}\right) \longrightarrow e^{-s^2/2}. \quad (34)$$

Thus the peak has height of order $k^{-1/2}$ and width of order $k^{-1/2}$, so its area has order k^{-1} , exactly as the mean requires. This also has a simple Poisson explanation: at the favorable phase one critical round has Poisson mean k , and $\mathbb{P}\{\text{Poisson}(k) = k\} \sim (2\pi k)^{-1/2}$.

Away from the moving peak, F_k can be exponentially smaller. The deepest phase lies asymptotically where the two nearest dyadic terms balance, at

$$u_0 = \log_2(2 \ln 2) = 0.471233627 \dots,$$

and

$$\min_{\theta} F_k(\theta) = k^{-1/2} \exp\{-\kappa k + O(1)\}, \quad \kappa = \ln 2 - 1 - \ln(\ln 2) = 0.059660101 \dots \quad (35)$$

This sharp peak and exponentially low trough are why the pure-phase picture eventually breaks down: more and more Fourier harmonics become visible.

9 Powers of two and other geometric subsequences

For powers of two, the exact formula can be written on a fixed dyadic grid. For example,

$$P_{2^m} = \frac{1}{2} \sum_{j \geq 1-m} 2^{-j} \left(1 - 2^{-m-j}\right)^{2^m-1}. \quad (36)$$

More generally,

$$H_{2^m,k} = (1 - 2^{-k}) \binom{2^m}{k} 2^{-mk} \sum_{j \geq -m} 2^{-kj} \left(1 - 2^{-m-j}\right)^{2^m-k}. \quad (37)$$

Termwise Poisson approximation, with the tail bounds used earlier, gives

$$\lim_{m \rightarrow \infty} H_{2^m,k} = F_k(0).$$

For any fixed $c > 0$ the same argument, or Theorem 4.1, gives

$$H_{\lfloor c2^m \rfloor, k} \longrightarrow F_k(\{\log_2 c\}). \quad (38)$$

Thus the natural convergent subsequences are obtained by fixing the position of n between consecutive powers of two.

The possible phases become arbitrarily fine in each interval $[2^m, 2^{m+1})$. For each fixed k , the set of subsequential limits of $(H_{n,k})$ is precisely the interval

$$F_k([0, 1]) = [\min F_k, \max F_k].$$

In particular,

$$\liminf_{n \rightarrow \infty} H_{n,k} = \min F_k, \quad \limsup_{n \rightarrow \infty} H_{n,k} = \max F_k.$$

10 Discussion

The apparent paradox is now resolved. The law of large numbers correctly describes the early stages: the audience is very close to being halved each round. It also correctly identifies $\log_2 n$ as the decisive time scale and suggests that doubling n shifts the process by one round. But the actual question of whether the chain visits a fixed k is decided only after the population has fallen to order one. At that point fluctuations are Poissonian rather than negligible. Because rounds occur at integer times, the fractional part of $\log_2 n$ survives and creates a log-periodic correction.

The relation $H_{2n,k} - H_{n,k} \rightarrow 0$ is therefore weaker than convergence. It says that the sequence is asymptotically invariant under doubling, not under arbitrary changes of n . As n moves between consecutive powers of two, $(H_{n,k})$ traces one copy of the profile F_k . For $k = 1$ that copy is so flat that its nonconvergence is almost invisible. As k grows, its moving peak has height of order $k^{-1/2}$, width of order $k^{-1/2}$, and phase $\{\log_2 k\}$ asymptotically; this is a simultaneous phase shift and concentration, not a pure translation.

This problem is also a basic randomized leader-election procedure. Its mathematical core—the uniqueness and multiplicity of the maximum of a geometric sample—has a substantial literature. In particular, Bruss and O’Cinneide proved the nonconvergence and logarithmically periodic asymptotic [4]; Kirschenhofer and Prodinger studied the full number of tied winners and its Fourier fluctuations [5]; Bruss and Grübel treated multiplicities for more general discrete distributions [6]; and Janson, Lavault, and Louchard explicitly studied the probability of passing through a prescribed fixed number of survivors in leader-election algorithms [8]. Thus the fixed- k problem and its log-periodic asymptotic are not new. The magician formulation, the direct occupation-time proof, and the phase-versus-shape discussion provide an expository route into the subject, but a publication should be positioned as an exposition or as a source of genuinely new extensions rather than claiming priority for the core theorem.

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